

**War of Operating Systems: The Paradigm Conquest of Analytic
Philosophy by Systems Ontology and the Rebirth of Philosophy**

**— Concurrently Discussing How Wittgenstein's Prohibition Became the Final
Tombstone of Old Philosophy**

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Abstract

Contemporary mainstream philosophy is not in crisis; it has completed its historical form: it has successfully transformed itself into an exquisite conceptual archaeology, whose ultimate achievement is proving its own 'inability to transform the real world'. This denouement reached the purity of logic and the thorough nihilism of practice in Ludwig Wittgenstein's prohibition of 'the unsayable'. He erected for it an impeccable tombstone.

Therefore, the work of this paper is not critique, but a declaration of death and a systematic autopsy, while simultaneously demonstrating the new civilizational operating system already erected upon the ruins of the old paradigm. By introducing the metalanguage of 'system dynamics', we establish the 'logical self-reference' feared by Wittgenstein as the source of systemic creativity (Axiom Zero), and reconstruct all the domains of existence, consciousness, and value that he deemed 'unsayable' into objects that can be spoken of, modeled, and designed.

This paper further demonstrates how this theoretical system accomplishes a fundamental reconfiguration of philosophy: its core object is elevated from trivial language games to the paradigm architecture of civilization, existence, and cognition; its core method leaps from critique and therapy to protocol design and system optimization; its core mission transforms from interpreting the world to designing the topological structure of civilization, eradicating systemic parasites, and guiding its phase transition toward 'metacognitive collective intelligence', with 'optimal individual ecological niche' as the ultimate criterion.

Finally, this paper declares: philosophy has not ended; it has been reborn in a 'war of operating systems'. The old operating system (represented by analytic philosophy) ceased evolution due to its kernel limitations; the new operating system (with Systems Ontology as its kernel) has already booted, enabling philosophy to regain its qualification as the 'first-principle architectonic science'—providing computable, executable foundational protocols and ultimate blueprints for civilization's future. This is not merely a theoretical victory, but a decisive leap in the cognitive hierarchy of civilization.

This paper is thus an escape from a cognitive hierarchy. It does not seek recognition within the old castle but provides the coordinates and landing protocols for a new continent. The rebirth of philosophy begins with rejecting its role as 'therapist' and embracing its destiny as 'architect of the civilizational operating system'.

A Quick Guide to Core Theoretical Concepts

Before delving into this intellectual confrontation, to prevent readers from getting lost in the dense terminology and concepts in the subsequent discussion, this quick-guide chapter is specially provided. It aims to offer you a concise "**mind map**", marking the most critical conceptual landmarks throughout the text. These definitions are not exhaustive academic expositions but are "**advance cognitive frameworks**" prepared to facilitate smooth reading.

1. Systems Ontology

This is the foundation of the entire theoretical edifice. It does not study systems in a specific domain (e.g., biological systems, economic systems), but investigates the most universal laws of how "**systems**" themselves come to exist, evolve, and interact. It attempts to answer: from elementary particles and living organisms to human societies and mathematical forms, do all these entities called "systems" share the same set of "rules of the game"? Its core proposition is: existence itself harbors an intrinsic "**self-organizing tendency**" (Meta-Axiom Zero), and any stably existing system embeds a set of "**constraint rules**" that manage itself and avoid collapse (Meta-Axiom One). In short, it is the ultimate physics and logic of "**how order emerges from potentiality and maintains itself**".

2. Noo-matter (Zhizhi)

This is a key concept for understanding human civilization. It refers to the other dimension of existence, relative to material entities, constituted by information, meaning, cognitive frameworks, and symbolic rules. Any complex system (a person, an organization, a culture) is simultaneously a "**material entity**" and a "**noo-matter pattern**". Your body is the material state, while your thoughts, memories, values, and the social norms you participate in constitute your noo-matter state. Civilization can thus be seen as a vast and complex **noo-matter ecosystem**.

3. Cognitive Framework

This is the "**operating system**" or "**filter**" through which each intelligent individual understands and intervenes in the world. It consists of your deeply held basic assumptions, value hierarchies, and thinking patterns. It determines what you see, what you ignore, and how you interpret events. When two people argue endlessly, it is often not because the facts differ, but because they are running different "**cognitive frameworks**". This theory emphasizes that there is no absolute "**God's-eye-view**" framework; all cognition is constrained by its framework (Axiom Three).

4. Logical Self-Reference

When a system reaches sufficient complexity, it acquires the ability to reflexively think about itself, much like a mirror that can not only reflect the external world but also reflect the mirror itself.

This "**system's pointing to and operation upon itself**" is logical self-reference. It is the root of consciousness, self-reflection, and creative intuition (Axiom Zero). Wittgenstein feared the logical paradoxes it brings, whereas we see it as the fundamental driving force enabling systems to self-renew and create.

5. Self-Organizing Tendency

This is the **"prime mover"** driving the universe from chaos to order. It is not purposeful design but an intrinsic, dynamical propensity for systems to spontaneously tend toward more ordered, more complex states. In the physical realm, it manifests as nebulae condensing into stars; in the biological realm, as life evolving from simplicity to complexity; in human society, as the enduring impulse to build civilization and create knowledge (Meta-Axiom Zero and Axiom Two). It is the universal source of **"negative entropy"** and the **"will to survive"**.

6. Noo-matter Ecological Niche

Borrowing from the biological concept of **"ecological niche"**, this refers to the unique position an individual occupies within the vast noo-matter ecosystem (society). This position is defined not only by your social profession but also by the cognitive skills, values, and interpersonal networks you master. A healthy civilization should enable each individual to find and optimize their unique **"noo-matter ecological niche"**, realizing their potential (Theorem Twenty-One).

7. System Topology Theory

This is the theory describing the macroscopic structure of civilization. It proposes that a healthy, sustainable civilization must, like a human being, possess two fundamental **"organs"**:

- **Pyramid Structure (Life Base):** Responsible for stability, efficiency, and survival guarantee, handling deterministic tasks (e.g., government, infrastructure).
- **Discrete Structure (Field of Consciousness):** Responsible for innovation, exploration, and meaning generation, handling non-deterministic tasks (e.g., academia, arts, R&D).

Civilizational health depends on whether these two can achieve dynamic balance through clear **"protocols"** (rather than brute command). Various civilizational pathologies (e.g., rigidity or chaos) can be diagnosed as dysfunctions in the relationship between these two structures.

8. Metacognition

That is, **"cognition about cognition"**. It is your ability to step outside your own **"cognitive framework"** and examine your own thought processes as an observer. It is the key psychological faculty for critical thinking, learning how to learn, and achieving **"cognitive escape"** (breaking through existing intellectual constraints). In this theory, it is the highest level in the three-tier model of consciousness (Theorem Twenty-Four).

9. Cognitive Escape

Refers to the process by which an individual, utilizing their **"metacognitive"** ability, breaks through the powerful gravitational pull of the dominant **"cognitive framework"** (or paradigm) of their society, achieving an intellectual leap. Its difficulty is directly proportional to the authority of the framework and the number of its adherents, and inversely proportional to the individual's own metacognitive ability (Theorem Twenty-Seven). All scientific and intellectual revolutions in history are successful **"cognitive escape"** events.

10. System Parasite

Refers to those subsystems within a social system that have deviated from their initial function of serving the whole (ultimately serving individuals) and instead take maximizing the extraction of system resources and maintaining their own privilege as their sole purpose. They consume societal

nourishment without creating corresponding value; they are the "**cancer cells**" of civilization (Theorem Twenty-Five).

Mastering these ten concepts equips you with the key to unlocking the intellectual world of this paper. They will serve as constellations, guiding your understanding of the subsequent critique of Wittgenstein, the redrawing of philosophy's territory, and the design of civilization's future. For more detailed, rigorous definitions and deductions, please refer to the Appendix: *Compendium of the Complete Theoretical System*.

Now, let us formally enter the battlefield of ideas.

(End of Quick Guide Chapter.)

This design aims to lower the entry barrier by presenting the most core, most prerequisite concepts in an accessible and structured manner. The appendix contains more complete axioms, theorems, and explanations of complex models for readers interested in deeper inquiry.

1. Precision Anatomy: The Source Code of Wittgenstein's Semantic Field and Its Systemic Translation

Before declaring war, one must precisely map the enemy's fortress and prohibited zones. What Wittgenstein left behind is not a set of randomly interpretable aphorisms but a logically stringent system of thought—a **"semantic operating system"** that draws absolute boundaries for philosophical discourse. To understand why we must transcend him, we must first understand him thoroughly enough to restate his fears and prohibitions in our own language.

1.1 Wittgenstein's "Primitive Operating System": The Ultimate Purge of the *Tractatus*

Let us return to the beginning for an intellectual archaeology. In the icy, self-disciplined text of the *Tractatus Logico-Philosophicus*, Wittgenstein performed one of the most thorough formatting operations in the history of philosophy.

- **The "Fact" Encoding of the World:** He declared, **"The world is the totality of facts (*Tatsachen*), not of things (*Dinge*)."** Here, "facts" are not events in everyday experience but units of existence in logical space. A fact is a "state of affairs" that makes an atomic proposition potentially true. The world was thus compiled into a static network of propositions woven by logical connectives—a clearly mappable structure.
- **The "Picture" Protocol of Language:** How can language describe the world? Because **"A proposition is a picture of reality."** This is not a metaphor but a strict logical isomorphism. Propositions and facts share a **"logical form"**, just as the grooves on a gramophone record share an abstract waveform structure with a symphony. The validity of language depends entirely on its ability to be a qualified "projection" of the world's logical structure.
- **Philosophy's "Delimitation" Task and Ultimate Silence:** Thus, the task of philosophy was defined with unprecedented clarity: **to draw a limit to thought**. On one side lies the "realm of facts," the abode of all propositions of natural science—the sayable. On the other side lies the unsayable. What is unsayable? Ethics, aesthetics, "the meaning of life," metaphysical propositions, and the feeling of "the world as a limited whole." Wittgenstein wrote with near-tragic resoluteness: **"Whereof one cannot speak, thereof one must be silent."** Philosophy, after completing its cleansing work, would itself be discarded—it is merely a ladder to be thrown away.

This is the first, and most solid, "cognitive framework" he established: the world is a static machine driven by logic and decomposable; language is its precise dashboard. Any data that cannot be displayed on this dashboard—those burning inquiries about value, meaning, and existence itself—were judged as system noise and **filtered out**.

1.2 The System's "Self-Debugging": The Turn and Therapeutic Retreat in the Philosophical Investigations

However, Wittgenstein later realized that the "logical crystal" operating system he built in his early years was too ideal and rigid, unable to accommodate the chaos and vitality of language's actual operation. Thus, he performed a major system upgrade, its core being the shift from the **"picture**

theory of meaning" to the "language-game" concept.

- **From "Picture" to "Game":** Meaning no longer stems from static correspondence with facts but resides in **use**. Words are like tools; their function is realized in specific "**language-games**." And language-games are rooted in broader "**forms of life**."
- **Philosophy as "Therapy":** If the meaning of language lies in specific use, whence do most traditional philosophical problems arise? Wittgenstein diagnosed that they arise from "**language going on holiday**"—words being abstracted from the specific rules of their games and operating autonomously in the air, thus producing dizzying but insoluble spasms like "What is time?" or "Where is the self?". Therefore, the philosopher's new task is not to construct theories but to become a **therapist**, bringing those lost words back to their "everyday home" by describing the actual use of language.

Seemingly moving from cold logic to warm life, this was actually a more profound retreat. Early on, he at least acknowledged a transcendent, logical "world" beyond language. Later, he simply contracted philosophy's territory to the observation and therapy of the application-layer phenomenon of "language-games." He abandoned any attempt to speak about the "**operating system**" **itself** (the foundations of existence, consciousness, value), instead declaring that all such inquiries were themselves "**diseases**" in need of therapy.

1.3 The Root of Fear: Philosophy Haunted by the Specter of "Logical Self-Reference"

Why did Wittgenstein impose such absolute prohibitions? What deep fear drove him from the silence of the *Tractatus* to the therapy of the *Investigations*?

This specter is "**logical self-reference**." When language attempts to speak about itself, when philosophy attempts to found the world-as-a-whole, it immediately falls into fatal circularities akin to the "Barber Paradox" or "Russell's Paradox." In Wittgenstein's view, this self-reference is a cancer of the language system, leading to the collapse of the entire edifice of meaning. His denial of the possibility of a "**private language**," his skepticism about describing "**inner processes**," all stem from this—any speech pointing to the inner self or the whole of existence cannot escape the trap of self-reference and is thus logically illegitimate.

Therefore, his entire philosophical project can be seen as an extremely cautious epidemic prevention campaign against the systemic crash risk posed by "logical self-reference." He demarcated "unsayable" prohibited zones, essentially establishing quarantine belts around the core of the language system to prevent the spread of the self-reference virus. His "therapy" was to eliminate any abstract word that might trigger a high fever of self-reference. This epidemic prevention system achieved unparalleled clarity but also rendered philosophy permanently immune-deficient against speaking about and constructing its own foundations, the civilizational whole, and the dynamics of existence.

1.4 Systemic Translation: Parsing the "Wittgenstein System" Within Our Framework

Now, let us perform a translation and diagnosis of Wittgenstein's philosophy using our own "**system dynamics**" language. This is not mere term substitution but running and analyzing his program on a more foundational operating system.

- **What is a "Language-Game"?** It is a **microscopic, local cognitive protocol**, a set of temporary rules evolved by a noo-matter community within a specific ecological niche to

coordinate action and maintain internal information homeostasis. It is essentially a microscopic manifestation of [**Axiom Three: The Relativity of Cognitive Frameworks**] at the group level.

- **The Dynamical Essence of "Meaning is Use":** In our view, the meaning of a symbol is its **functional utility** as a noo-matter gene in participating in the **"self-organizing tendency"** ([Axiom Two]) practice of a system (individual or community). Wittgenstein saw the surface phenomenon of "use" but failed to penetrate to its underlying driving force—the system's will to survive and expand.
- **What is the "Self-Reference" He Feared?** It is precisely the inevitable attribute of systems reaching a critical complexity described by our [**Axiom Zero: The Dynamical Principle of Logical Self-Reference**]. He saw this as a fire to be extinguished; we see it as the **"intrinsic energy source"** enabling systems to introspect, iterate, and creatively leap.
- **Was His Diagnosis of "Philosophical Disease" Correct?** Partially correct. When a "language-game" (local protocol) is misapplied to global questions, mismatches and confusion indeed arise. But his error was in believing that all global, foundational questions were **illusions** produced by such "mismatch." He failed to recognize that there exists a higher-level **"meta-game"** about how protocols themselves are generated and evolve—and this is precisely the domain where philosophy should reside.

Diagnostic Conclusion: Wittgenstein constructed an expert system exceptionally adept at handling the compatibility and bugs of **"application software"** (everyday language, specific games). But he wrote an ultimate directive for this system: **strictly prohibit probing, modifying, or rewriting the "operating system" itself** (the dynamical foundations of existence, consciousness, value). He misdiagnosed all the operating system's background processes as viruses needing eradication. It is precisely this directive that caused subsequent philosophy, while gaining precision, to permanently lose system-level architectonic and transformative power. And we are about to demonstrate that our theory, by positively accepting and managing the "self-reference" he feared, has acquired the permissions to rewrite the operating system.

1.5 Systemic Translation: Wittgenstein's Unfinished Project

"Wittgenstein's description of the complexity of 'language-games' is unprecedentedly precise; he correctly pointed out the fluidity of meaning in usage.

However, his toolbox contained only two wrenches: **'description'** and **'therapy'**. He could exquisitely describe how every part of an engine rattles (language-games) and suggest making the rattling stop (returning to everyday use), but he refused to ask and was powerless to answer: What is the design principle of this engine? Where should it drive civilization? His philosophy thus became a custodial art for a **'phenomenological museum'**."

Systemic Translation (Presented in a Table):

Wittgenstein's Observation (Phenomenal Layer)	Systems Ontology Translation (Dynamical Layer)	Theoretical Components
"Language-games" and their rules	Micro-cognitive protocols , temporary rule convergence formed to reduce interaction costs ([Theorem Seventeen]) in specific contexts.	Axiom Three (Framework Relativity), Connection Tendency Formula
"Meaning is use"	The " meaning " of a symbol as a noo-matter gene ([Axiom Four]) is its functional utility in participating in the " self-organizing tendency " ([Axiom Two]) practice of a system.	Axiom Two, Axiom Four, Noo-matter Gene Propagation Equation
"Philosophical disease" (obsession with the whole)	Cognitive framework mismatch resulting from misapplying micro-protocols designed for local problems to macroscopic system architecture problems.	System Topology Theory (Structure-Mode Matching Principle)
"Whereof one cannot speak..."	A " protocol incompatibility " error inevitably triggered when the old "semantic reference" language-game attempts to handle Systems Ontology problems. Not that the object is mysterious, but the tool is low-dimensional.	Meta-Axiom One (Systemic Constraint Dissolves Its Legitimacy)

Conclusion: "Therefore, we are not 'simplifying' or 'refuting' Wittgenstein. **We are completing him.** We accept all the phenomenal facts he delineated but replace the underlying operating system, thereby enabling the execution of programs about existence and civilization itself—programs his system could not process. His prohibition was an '**insufficient memory**' prompt of his system, not a truth about the world."

2. Frontal Conquest: The Rise of Systems Ontology as the Meta-Operating System

The anatomy is complete; the diagnosis is settled. Wittgenstein's philosophical system is like a closed castle designed to withstand logical storms, its internal order impeccable at the cost of forever losing the ability to survey and shape the vast external world. Now, it is time to demonstrate that we are not building a more splendid palace outside the castle, but providing an entire set of "**dynamical foundations**" and "**system architecture**", enabling the castle itself to be safely deconstructed, reconfigured, and elevated into an organic life-form capable of self-growth, regulation, and design. This chapter unfolds the frontal conquest. We will prove that Systems Ontology is not another "philosophical viewpoint," but a more universal **meta-operating system**. It can precisely describe how Wittgenstein's "language-games" operate as subsystems, and more importantly, it can process the "foundations of existence" and the "civilizational whole" that he prohibited. Ultimately, it rewrites philosophy's mission from "treating language diseases" to "**designing civilizational protocols.**"

2.1 New Language: From Depicting “Language-Games” to Modeling “System Dynamics”

Wittgenstein’s later philosophy is, in essence, a form of phenomenological description. He was like an immensely patient naturalist, meticulously classifying and displaying specimens of various “language-games,” telling us: “Look, this is how it is used.” This description brought unprecedented clarity but also stopped at clarity. It could not answer: Why do these games emerge? How do they compete and evolve? What is their underlying driving force?

Our system introduces “**system dynamics**” as the new metalanguage. This is not merely a change of terminology but a leap in the level of thinking.

- **Dynamical Modeling of “Language-Games”:** A “language-game,” in our view, is a temporary **cognitive collaborative protocol**, a spontaneous rule convergence formed by multiple noo-matter systems (individuals) to reduce interaction costs in specific contexts ([Theorem Seventeen: Social Interaction Energy Cost Theorem]). We can model the formation probability of such a gaming community using the connection tendency score. Its formula precisely captures the driving forces and resistances of connection establishment:

$$S = \frac{W_A \times W_B}{D^2 + C}$$

Here, W_A, W_B represent the **will-to-survive intensity** of the interacting parties, D is the **cognitive framework distance** between them, and C is the **cognitive inertia constant**. This formula shows that connection establishment requires not only mutual willingness (numerator) but is also constrained by the **amplifying effect** of cognitive differences (denominator), thereby quantifying the communication chasm caused by differences in “forms of life” as described by Wittgenstein.

Further, we can use the noo-matter gene propagation equation to simulate how a concept, narrative, or rule spreads, mutates, and gains dominance within a game network, dynamically describing the evolution of the “language-game” itself:

$$\frac{dP(t)}{dt} = \beta \cdot I \cdot C \cdot P(t) \cdot [1 - P(t)] \cdot N - \delta P(t) - \gamma \cdot P(t) \cdot Q(t)$$

Here, $P(t)$ is the meme prevalence, β the infection intensity, I the influencer distribution, C the framework compatibility, N the network effect, δ the forgetting rate, γ the competition inhibition coefficient, and $Q(t)$ the competing meme prevalence. This equation treats the propagation of symbols as a dynamical competition within the cognitive ecology, surpassing Wittgenstein’s static “description of use” and revealing how “meaning” dynamically emerges, shifts, and vanishes at the **group** level.

Wittgenstein stopped at ‘describing use’; we can simulate the dynamical processes of its generation, propagation, and evolution. His work was like drawing a topographic map of language; we have written the climatology and tectonic equations that shape this terrain.

- **Legitimate Speech in the “Unsayable” Domains:** The language of system dynamics, because it deals with relations, processes, and patterns rather than directly referring to “mysterious objects,” circumvents Wittgenstein’s prohibition against “referential metaphysics.” We do not say “what the world-as-a-whole is”; we describe the **“eternal dynamical cycle driven by the self-organizing tendency”** ([Meta-Theorem One]). We do not define the “self”; we analyze **“how logical self-reference emerges as the metacognitive interface within the three-tier consciousness model”** ([Theorem Twenty-Four]). We

transform his “unsayables” into sayable, calculable system behaviors. This means our language can not only cover his entire descriptive domain but also delve beneath to reveal the **driving mechanisms** behind these phenomena and expand upward to speak about his prohibited zones. **His language is the map; our language is the projection laws and geodynamics for drawing maps.**

2.2 New Objects: From Debugging “Application Software” to Reconstructing the “Operating System”

This is the core of the conquest. Wittgenstein reduced almost all philosophical problems to misuses of “language-games,” i.e., logical conflicts or conceptual confusions at the application software level. His “therapy” was an excellent debugging tool, capable of resolving many program crashes (philosophical perplexities). However, when an entire civilizational system falls into chronic paralysis of “**transformative impotence**”—as Marx denounced as “only interpreting the world,” Mills lamented as “detachment from reality,” Habermas critiqued as “system colonizing the lifeworld”—debugging individual applications is futile. **The problem lies with the operating system itself.**

The fundamental revolution of our theory lies in re-anchoring philosophy’s core object at the **paradigm-architectural level of existence:**

- **Existence Itself:** No longer a mysterious background, but expressible as the “**self-organizing tendency**” of [Meta-Axiom Zero], the ‘prime mover’ of all negative entropy, structuration, and will-to-survive phenomena in the universe.
- **Consciousness and Cognition:** No longer private illusions or linguistic epiphenomena, but emergent properties of [Axiom Zero: The Dynamical Principle of Logical Self-Reference] in complex systems, their internal structure clearly parsable by the “**hierarchical collaborative model**” of [Theorem Twenty-Four].
- **Value and Meaning:** No longer unsayable ethics, but the “**system efficacy function of sustaining component collaboration**” defined by [Axiom Five], whose **concrete instantiation** in human society is the “**optimal individual ecological niche**” of [Theorem Twenty-One].
- **Civilization and Society:** No longer an ineffable chaos, but the **super-system** depicted by [System Topology Theory], dynamically **coupled** through ‘protocol modes’ by the ‘pyramid structure’ (life base) and the ‘discrete structure’ (field of consciousness).

We proclaim: the primary task of philosophy is not debugging the ‘language’ application, but diagnosing and upgrading the ‘civilization’ operating system. The consensus pathology of ‘transformative impotence’ in the socio-philosophical field is the direct consequence of the old philosophical paradigm (of which the Wittgensteinian tradition is the most radical form) self-limiting and refusing to touch the operating system. They were busy performing increasingly **fine-grained syntactic checks** at the application layer, while turning a blind eye to system-level resource monopolies (pyramid hypertrophy), process rigidity (cognitive framework ossification), and viral parasites ([Theorem Twenty-Five] system parasites), or dismissing them as ‘non-philosophical problems’.

2.3 New Method: Managing “Self-Reference,” Not Fearing It—From Therapy to Design

The terminus of Wittgenstein’s method is “**therapy**,” i.e., eliminating intellectual spasms caused by

self-reference and transgression, letting everything return to “calm.” Our starting point is precisely **accepting and disciplining “self-reference.”**

- **Rehabilitating “Logical Self-Reference”:** [Axiom Zero] establishes it as the **creative source** of complex systems. Wisdom, reflection, and paradigm revolutions are all rooted here.
- **Establishing “Systemic Constraints”:** [Meta-Axiom One] then provides the “**constitutional principle**” for managing self-reference. Systems naturally **dissolve** the legitimacy of “**malignant self-reference**” leading to their own collapse (e.g., logical paradoxes), while encouraging **constructive self-reference** (e.g., scientific theory self-correction). This is equivalent to setting up a kernel protection mechanism for the operating system, enabling deep self-iteration without falling into infinite recursive crash.
- **Methodological Leap: From Therapy to Design:** Therefore, our core method is no longer treating individual thinkers’ language diseases but **designing robust protocols for civilizational systems.**
 - We design “**truth discovery protocols**” to efficiently **screen and select cognitive genes** within the discrete field of consciousness, replacing authoritarian dogmatism.
 - We design “**attention auction protocols**” to optimize the allocation of civilizational innovation resources via market mechanisms, replacing bureaucratic planning or chaotic competition.
 - We design “**value alignment protocols**” to ensure any system-level decision can be traced back to promoting individual development ([Axiom Five]), preventing system **alienation**.
 - We define the “**civilization state function**”, providing a quantifiable core indicator for diagnosing system health.

$$\text{Civilization Potential} = \left(\frac{1}{n} \sum_{i=1}^n \text{INE}_i \right) \times \text{SRE} \times (1 - \text{SPB})$$

Where: INE = Individual Niche Realization, SRE = System Resource Fluency, SPB = System Parasite Burden.

This formula itself is a powerful diagnostic tool, transforming vague critiques (e.g., “detachment from reality”) into measurable, comparable, and intervenable system variables.

This marks a fundamental shift in philosophy’s role: from a gardener clearing weeds in the backyard of thought to an **architect and engineer** designing irrigation systems, planning ecological layouts, and cultivating new species on the civilizational front lines. **We are no longer content with making the world ‘look’ clear; we want the world to run and evolve efficiently toward the direction of ‘optimal individual ecological niche’.**

2.4 Interim Victory: Control of the Operating System Has Been Transferred

Through the frontal assault on these three dimensions, we have already achieved transcendence over the Wittgensteinian paradigm. This is not supplementation but overlay; not dialogue but upgrade.

We have proven:

- Our language can explain his entire world, and more deeply.
- Our objects encompass and surpass his prohibited zones, targeting the core of civilizational architecture.

- Our method resolves the root of his fear (self-reference) and transforms it into a constructive force.

Therefore, the discourse and control of philosophy have historically shifted from the hands of the “**language therapist**” to the hands of the “**system architect**”. The old operating system warned us “**do not touch the kernel**”; the new operating system proclaims: “**The kernel is open source, its dynamical laws have been mastered; now, we can and must write more elegant code for civilization’s future.**”

This frontal conquest has reclaimed for philosophy the grand vision and transformative power it once lost. In the next chapter, we will specifically demonstrate how this new operating system rebuilds solid and magnificent system modules in each key domain exiled by the old paradigm—existence, consciousness, value, and civilization.

3. Reclaiming Lost Territory: Rebuilding the Kingdom of Philosophy at the Operating System Level

The bugle of conquest has been sounded, and the forbidden zones demarcated by the old operating system are crumbling. The Wittgensteinian tradition regarded holistic discourse on existence, consciousness, value, and civilization as a “language disease” requiring therapy, exiling them beyond the borders of philosophy. This directly led to thought’s “impotence to transform reality.” Now, it is time to use our newly architected operating system to rebuild solid, operational, and even optimizable system modules upon these ruins. This chapter will demonstrate how we transform the mysterious “unsayable” domains of the old paradigm into the “sayable, analyzable, designable” core components of the new paradigm, thereby thoroughly reclaiming philosophy’s lost kingdom.

3.1 Module One: Existence — From Silent “Mystery” to Driving “Self-Organizing Tendency”

At the conclusion of the *Tractatus*, Wittgenstein relegated the feeling of “the world as a limited whole” to the unsayable realm of mystery, deeming any speech about existence itself a logical transgression. This directly caused philosophy’s collective aphasia regarding ontological questions, ceding them to theology, poetry, or silent experience.

Our Systems Ontology, centered on [**Meta-Axiom Zero: The Principle of Self-Organizing Tendency**], directly responds to this prohibition. We no longer ask the static, essentialist question “What is existence?” which easily falls into linguistic circularity, but describe the dynamical process of “**how existence operates.**”

- **The Dynamical Definition of Existence:** Existence is an eternal cyclical process driven by the “self-organizing tendency” and regulated by “systemic constraints” ([Meta-Theorem One]). Its manifestation is: potential domain → system emergence → system evolution → system collapse → return to potential domain. This cycle has no external purpose; its “meaning” is the process itself.
- **Legitimate Speech about the “Unsayable”:** We do not directly define the “world-as-a-whole” but describe the fundamental **motivational pattern** through which the world-as-a-whole presents itself and evolves. This is akin to not defining “life” but describing life

processes such as metabolism, irritability, and reproduction. The self-organizing tendency is the “metabolism” of existence, the first-principle **driving force** behind negative entropy, structuration, and will phenomena.

- **From Mystery to Operation:** Understanding existence as a tendency and process makes it **actionable**. We can study the critical conditions for system emergence ([Theorem Eight]), analyze which constraints maintain system health, and even contemplate how to guide a specific form of existence—civilization—toward a higher-order phase transition to “metacognitive collective intelligence.” Existence transforms from a static idol to be **gazed upon** into a “cosmic engineering” object whose dynamical processes we can participate in.

3.2 Module Two: Consciousness and Self — From “Limit of the World” to “Dynamical Interface of Logical Self-Reference”

Wittgenstein famously held that “The subject does not belong to the world: rather, it is a limit of the world” (*Tractatus* 5.632), and that the “self” is a philosophical illusion arising from a misunderstanding of the grammatical form “having experience.” He thoroughly expelled consciousness and self from the sayable world.

Our system, through [**Axiom Zero: The Dynamical Principle of Logical Self-Reference**] and [**Theorem Twenty-Four: The Hierarchical Collaborative Model of Consciousness Phenomena**], not only welcomes consciousness back into the realm of analysis but further parses it as the most sophisticated metacognitive interface within the civilizational operating system.

- **The Demystification of Consciousness:** Consciousness is no longer a ghost but the “**self-referential**” capacity that emerges from the internal information-processing of any logical system reaching a specific complexity. It allows the system to cognize and operate upon itself as an object, thereby generating an internal “logical sub-universe.” Human consciousness, creative intuition, and reflective ability are the highly developed forms of this dynamical principle in carbon-based neural networks.
- **The “Self” as System Interface:** [Theorem Twenty-Four] deconstructs subjective experience into a three-tier collaborative resonance architecture: “Biological Instruction Layer – Subconscious Processing Layer – Metacognitive Layer.” The so-called “**sense of self**” is precisely the experience of the “system console” presented when the metacognitive layer, as the system’s highest coordination center, integrates information from lower layers and forms decision-making consensus (high alignment). It is not an illusion but the phenomenological presentation of the system’s self-referential function.
- **Providing a Cognitive Science Foundation for Philosophical Critique:** This model explains why Wittgenstein’s “therapy” is sometimes effective—when philosophical perplexity stems from the metacognitive layer’s failure to integrate chaotic signals from lower layers (especially the subconscious layer) resulting in low alignment, clear linguistic description (symbolic **sorting** by the metacognitive layer) can indeed have a “therapeutic” effect. But it further indicates that much profound suffering stems from **structural conflicts between layers** ([Theorem Forty-One]), far beyond what language therapy can solve, requiring systematic cognitive framework reconstruction or ecological niche adjustment. We transform consciousness from an expelled “trouble” into a “core system process” that can be debugged, optimized, and even upgraded.

3.3 Module Three: Value and Meaning — From “The Silence of Ethics” to “The System’s Criterion of Value”

This is the domain of Wittgenstein’s most severe prohibition: “Ethics is transcendental. Ethics cannot be put into words.” (*Tractatus* 6.421). Questions like “the meaning of life” were typical cases of language misuse. This attitude forced philosophy into silence or indefinite conceptual postponement when facing the most burning **practical** issues of social justice and life direction.

Our system, with [Axiom Five: The Criterion of System Value] and its corollary [Theorem Twenty-One: The Theorem of Optimal Individual Ecological Niche], constructs a theory of value naturally grown from system dynamics, **shattering this silence completely**.

- **The System-Genetic Theory of Value:** Value is not a mysterious transcendental command but a functional requirement for a system to maintain its own existence. The continued existence of any complex system depends on its ability to consistently meet the specific needs of its internal components, thereby sustaining collaboration ([Axiom Five]). Value is the system’s efficacy in responding to its components’ needs.
- **The Ultimate Formula for Human Societal Value:** For the specific system of human civilization, its ultimate criterion of value stems from the “**original protocol**”—individuals collaborate by ceding some rights to form a system to enhance survival efficacy. Therefore, the ultimate value and legitimacy of any social system lies solely in the degree to which it promotes the survival and developmental potential of *each and every* individual within the system ([Theorem Twenty-One]). This theorem transforms “humanism” from a vague moral slogan into a clear, analyzable core indicator of system health.
- **The Reconstruction of “Life’s Meaning”:** Based on this, [Theorem Nine] reconstructs the pseudo-question “What is the meaning of life?” into the true question: “**How does a discrete node (a human being) dynamically define itself in the continuous resistance against systemic pressure and the active construction of collaborative networks?**” Meaning is no longer about finding a ready-made answer but is the practical process itself of the will-to-survive carving a path toward “optimal individual ecological niche” within the system topology.
- ****Connecting “Is” and “Ought”:** Our theory accomplishes a crucial connection: deriving the *ought* direction of civilizational design from the *is* fact of system dynamics (the self-organizing tendency drives system maintenance and optimization). Value judgments can henceforth be based on empirical analysis concerning system health and individual development, rather than floating upon unsayable intuition or authority.

3.4 Module Four: Civilization and History — From “Non-Philosophical Objects” to “Designable System Topology”

The analytic tradition viewed civilization, history, and social structures as objects for concrete sciences (history, sociology), with philosophy providing at most methodological reflection. The Wittgensteinian tradition reinforced this segregation. This led to philosophy’s collective incapacity before grand historical narratives and civilizational design, the most **direct** manifestation of socio-philosophy’s “impotence to transform.”

Our [System Topology Theory] and [Civilization Dynamics] are precisely the super-modules constructed to reclaim this ultimate lost territory. We define civilization directly as philosophy’s core object of operation.

- **The Topological Definition of Civilization:** Civilization is a noo-matter ecosystem based on shared meta-narratives and collaboration protocols, capable of achieving dynamic balance between a “Pyramid Structure” (Life Base) and a “Discrete Structure” (Field of Consciousness). Its essence is a specific **system topology configuration**.
- **The Science of Diagnosis and Design:** Based on this definition, we develop a complete language for diagnosis and design:
 - **Diagnostic Tool:** The **[Civilization State Function]** Civilization Potential = $\Sigma(\text{Individual Niche Realization}) \times \text{System Resource Fluency} \times (1 - \text{System Parasite Burden})$, transforms civilizational health into a quantifiable, comparable formula.
 - **Pathology: [System Topology Theory]** precisely defines civilizational diseases such as “Pyramid Hypertrophy” (innovation suffocation), “Discrete Field Disorder” (system disintegration), and “Interface Canceration” (resource misallocation).
 - **Design Blueprint:** Proposes a civilizational upgrade architecture centered on the “**Metacognitive Interface**” and “**Advanced Protocol Modes**” (Truth Discovery, Attention Auction, Value Alignment), aimed at guiding civilization’s phase transition toward “Metacognitive Collective Intelligence.”
- **Philosophy as Civilizational Architect:** This means philosophy is no longer merely a commentator on history or a critic of society, but should become civilization’s **diagnostic physician and architectural engineer**. Its task is to take civilization’s pulse, prescribe remedies, and directly participate in developing the protocol layer of the new operating system. This is the most **thorough response** to Marx’s call to “change the world”—not through violent revolution, but through systematic cognitive reconstruction and protocol design.

3.5 Mathematical Models: A Philosophical Declaration — Not Description, but Protocol

- **The Old Paradigm’s View of Mathematics:** In the old paradigm, mathematical formulas in philosophy were descriptive, attempting to “depict” pre-existing social laws using mathematical language. When data was lacking, they were dismissed as fantasy.
- **The New Paradigm’s View of Mathematics:** In Systems Ontology, these equations are **normative sets of protocol instructions**. Their primary purpose is not to define “what the world is,” but “what dynamical rules we should follow if we wish to construct a healthy system.”
- **Example 1 (Connection Tendency Formula):** It is not a law predicting “how people must interact,” but a screening protocol for optimizing connection efficiency when constructing a collaborative cognitive network. It is an algorithmic principle for designing social platforms, not a pallid description of the human status quo.
- **Example 2 (Civilization State Function):** It is not merely a measuring stick for the strength of existing civilizations (though it can be used for diagnosis), but the **mathematical definition of the ultimate value goal toward which civilization should evolve**. It is the objective function in a design blueprint.
- **Conclusion:** “Therefore, questioning these equations for ‘lacking empirical data’ is like questioning a newly drafted constitution for ‘not yet being historically tested.’ They are not reports on the old world but engineering specifications for building a new one. Their

verification lies in whether systems built according to these specifications are healthier and more creative. We invite the world to undertake this great engineering experiment.”

Epilogue: Territory Reclaimed, Kingdom Reborn

Thus, we have completed a systematic “reclamation of territory.” All “unsayable” forbidden zones demarcated by the Wittgensteinian tradition have been transformed, within our new operating system—Systems Ontology and Noo-matter Ecology—into sayable, analyzable, calculable, designable core functional modules.

The Old Paradigm’s “Forbidden Zone” (Illusion/Disease)	The New Paradigm’s “Core Module” (Operable Object)	Key Theoretical Components
Existence Itself	The Driving “Self-Organizing Tendency”	Meta-Axiom Zero, Meta-Theorem One
Consciousness & Self	The Dynamical Interface of “Logical Self-Reference”	Axiom Zero, Theorem Twenty-Four
Value & Ethics	The System’s “Criterion of Value” & “Optimal Individual Niche”	Axiom Five, Theorem Twenty-One, Theorem Nine
Civilization & History	The Diagnosable & Designable “System Topology”	System Topology Theory, Civilization State Function, Phase Transition Model

Philosophy, therefore, has been reborn from a “therapist” patrolling language’s byways into a “system architect” commanding the vast continent of civilization. It has reclaimed its lost kingdom, and this kingdom, for the first time, possesses precise mapping tools, dynamic evolution models, and conscious design blueprints.

The old operating system locked its upgrade path due to kernel fear; the new operating system opens the possibility of infinite evolution due to its kernel’s strength and security. This is not the end of philosophy, but philosophy’s true beginning at a higher dimension. In the next chapter, we will look forward to the **brand-new future** opened by this paradigm revolution: the mission and prospects of philosophy as the **“first-principle design science.”**

4. The Rebirth of Philosophy: As First-Principle Design Science

The campaign is concluded; the territory is charted. We have systematically reclaimed the forbidden zones of the old philosophical operating system, reconstructing existence, consciousness, value, and civilization as operable modules. Standing now on the watchtower of this new continent, we must answer an ultimate question: After reclaiming lost territory, what must philosophy become?

The answer has emerged spontaneously from the unfolding of the theory: philosophy must, and will, be reborn as **“First-Principle Design Science.”** This is not a self-aggrandizing claim of disciplinary status, but the inevitable destination of philosophy when it re-anchors itself, upon the foundation of Systems Ontology, to the universal laws of system existence—from elementary particles to civilizational architecture. When philosophy grounds itself in the dynamical laws of existence itself, it can no longer be content with being a footnote to history, a mirror of reality, or a doctor of language; it

must become the architect of the future, the writer of protocols, the navigator of evolution.

4.1 From “Interpreting the World” to “Designing Evolution”: The Ultimate Upgrade of Philosophy’s Mission

The roar Marx issued a century and a half ago—“The philosophers have only *interpreted* the world, in various ways; the point, however, is to *change* it”—has remained an unhealed scar, relentlessly interrogating philosophy’s practical character. The analytic tradition, especially Wittgenstein’s turn, responded to this interrogation in a near self-mutilating manner: since direct speech aiming to “change the world” easily lapses into metaphysical nonsense, we will only perform “therapy” to ensure clarity of thought, leaving the task of changing the world to others. This cowardly retreat directly resulted in philosophy’s “aphasia” and “impotence” before reality.

Our theoretical system, by decomposing the macroscopic demand to “change the world” into a series of designable, iterable, verifiable systems engineering projects through Systems Ontology, has finally found a legitimate path for philosophy to fulfill its transformative mission.

- **Design, Not Speculation:** The core output of the new philosophy is no longer metaphysical systems or critical theories, but a series of foundational protocols and architectural blueprints based on system dynamics. For instance, the “Truth Discovery Protocol” is an institutional design for how academic communities efficiently screen and select knowledge; the “Value Alignment Protocol” is a constitutional principle ensuring that any social organization does not deviate from the development of its members; “System Topology Theory” is an architectural guide for how civilization balances stability and innovation. These design products, like open-source software protocols or urban planning statutes, possess a **strong capacity for real-world intervention**.
- **Evolution, Not Revolution:** We do not advocate violently overlaying existing civilization with a utopian blueprint. On the contrary, guided by the hierarchical emergence and evolution theorems of Systems Ontology, we view civilization as a running complex system with path dependence. Philosophy as design science has the task of diagnosing the current system’s health (using the Civilization State Function), identifying and suppressing system parasites (applying Theorem Twenty-Five), designing and piloting superior local protocols (e.g., new governance rules for collaborative platforms), and guiding the system toward a gradual phase transition to a higher-order “Metacognitive Collective Intelligence.” This is a patient “evolutionary engineering” based on system dynamics.

Thus, philosophy’s mission is upgraded from “interpreting the world” (old metaphysics) and “treating language” (analytic philosophy) to **“designing civilizational evolution based on system laws.”** It is no longer a cleaner following history’s carriage but an engineer drawing the track map for the next leg of the civilizational train.

4.2 The Meaning of “First-Principle”: Systems Ontology as Metascience

What is “First-Principle Design Science”? Here, “first-principle” does not denote supreme importance but foundational and prerequisite nature. Systems Ontology itself is first-principle; it reveals the universal dynamical laws from particles to minds to civilizations. Philosophy rebuilt upon this foundation naturally becomes a **“metascience,”** providing a unified paradigmatic foundation and design grammar for all social sciences, humanities, and future sciences involving complex systems, cognition, and value.

- **A Unified Conceptual Framework:** Concepts such as matter/noo-matter duality, logical self-reference, self-organizing tendency, cognitive framework, and system value provide a commensurable metalanguage for economics (studying the flow of noo-matter resources), political science (studying system topology and power), psychology (studying consciousness-layer conflicts), and pedagogy (studying cognitive protocol alignment). Disciplines are no longer isolated islands of discourse but mark different regions on the same “system map.”
- **The Meta-Method of Design:** We provide not specific social policies or educational curricula, but the meta-methods required for designing such policies and curricula. For example, the “Cognitive Interaction Effectiveness Theorem” provides evaluable dimensions for any educational or communication design:

$$EI = P \times C \times (1 + E)$$

where EI is the Effectiveness Index, P is Protocol Matching, C is Pattern Completeness, and E is Energy Directionality. This quantitative framework makes previously vague instructional design measurable and optimizable.

Similarly, the “Connection Tendency Formula” provides a tool for interpersonal dynamics analysis in organizational management and team building:

$$S = \frac{W_A \times W_B}{D^2 + C}$$

Experts in various fields, applying these meta-methods derived from Systems Ontology combined with their own knowledge, can produce more effective practical solutions.

- **The Ultimate Anchor of Value:** The “Optimal Individual Ecological Niche” established by [Theorem Twenty-One] provides a unified ultimate criterion of value for all specific domain designs. An economic policy, a law, an education system—its merit can be traced back to a fundamental question: To what extent does it promote, rather than inhibit, the realization of each unique individual’s potential within the system? This provides a compass to prevent designs in various fields from falling into local optima at the expense of overall health.

In this sense, philosophy with Systems Ontology as its kernel is reborn as the foundation, blueprint, and compass for all “design sciences” oriented toward human affairs. It does not replace specific disciplines but makes their synergy possible and ensures their direction aligns with civilization’s ultimate health.

4.3 Addressing Critiques: This Is Not Utopianism, But the Metatheory of Systems Engineering

This grand proclamation will inevitably invite skepticism. The sharpest critique may be: Is this not another philosopher’s arrogant “rational constructivism”? How is it different from the failed utopian designs of history?

Our answer is clear: **Fundamentally different.** The distinction lies precisely in our theoretical kernel—the dynamics and complexity thinking inherent in Systems Ontology.

- **Opposing “Omnipotent Reason,” Embracing “Evolutionary Trial and Error”:** We are deeply aware of the inherent unpredictability of complex systems. Therefore, our “design” is essentially about setting initial conditions, establishing feedback mechanisms, and cultivating evolutionary environments, not drawing a final blueprint that must be fully realized. The “protocols” we design are more like evolvable algorithms, whose concrete forms in practice

will be determined by the emergent interactions of participants.

- **Embedded “Self-Critique” and “Cognitive Escape” Mechanisms:** Our theoretical system itself embeds [Theorem Twenty-Seven: Cognitive Escape Theorem] and [Axiom Three: The Relativity of Cognitive Frameworks]. This means any design guided by it must contain channels for its own critique, revision, and transcendence. It encourages, rather than suppresses, the emergence of “metacognitive interfaces,” thereby avoiding **rigidification into new dogma**.
- **Using “System Health” as the Criterion, Not “Abstract Ideals”:** Utopias often serve the extreme realization of an abstract ideal (equality, freedom, efficiency). We use the comprehensive health indicator of the **“Civilization State Function”** as our measure:

$$\text{Civilization Potential} = \left(\frac{1}{n} \sum_{i=1}^n \text{INE}_i \right) \times \text{SRE} \times (1 - \text{SPB})$$

It demands dynamic balance across multiple dimensions: individual realization, resource fluency, and system integrity. This is essentially a pragmatic wisdom that opposes any extremism, emphasizing systemic resilience.

Therefore, what we provide is not a static map of an “ideal republic,” but a **“dynamic navigation science”** for shipbuilding, navigation, warning, and repair while sailing the complex seas. It does not promise heaven but promises an operating system that can make our civilization-ship more storm-resistant, better directed, and more capable of allowing every crew member to utilize their strengths.

4.4 Conclusion: The War of Operating Systems Is Over; The Rebirth of Philosophy Has Just Begun

Let us return to the starting point of this “war.” Wittgenstein, with his unparalleled rigor, wrote the epitaph for the old philosophical paradigm: philosophy is either the clarification of logic or the therapy of language; beyond that lies delusion. He closed the door to grand narratives and direct practice, believing this to be the ultimate defense of philosophy’s dignity.

We have demonstrated through this intellectual campaign that he was defending an already impoverished castle. The true dignity of philosophy lies in confronting the dynamical laws of existence itself and using the most powerful tools available to reason—systems science, dynamical modeling, axiomatic deduction—to analyze them, speak of them, and thereby design their evolutionary paths toward better states.

Systems Ontology, as the first-principle revealing the laws of all systems, has successfully upgraded philosophy from a self-enclosed castle to an open launch platform facing the starry sea of civilization. The old operating system ceased updates out of fear of self-reference and the whole; the new operating system gains infinite upgrade potential by mastering self-reference and **comprehending the whole in its essence**.

Therefore, we must solemnly declare:

The old paradigm that reached its peak and end in Wittgenstein—confining philosophy to linguistic analysis and conceptual therapy—has fulfilled its historical mission and, due to its self-castration of “transformative power,” has come to an end.

Simultaneously, a new philosophical paradigm has been born, with Systems Ontology as its kernel and the mission of designing civilizational evolution and promoting individual realization.

Philosophy has not died. It has merely, after a long slumber and self-doubt, completed a difficult “cognitive escape,” breaking free from the gravitational field of the old paradigm and transitioning into a broader, more responsible new orbit. Here, it is no longer a passive interpreter or therapist, but an active designer and architect.

The war of operating systems is over. The rebirth of philosophy has just begun. The future task is arduous and concrete: to transform this blueprint of a meta-operating system into executable protocols, embedded into the fabric of our civilization. This requires no longer the solitary contemplation of genius, but the collective construction of a collaborative cognitive network. And this is precisely the practice called for and destined to be engendered by the new philosophical paradigm.

Postscript: The Ultimate Response: A Guide to Cognitive Framework Switching for the Potential Reader

On How to Read This Paper: A Guide for Switching Cognitive Protocols

If you feel intense discomfort while reading, find the content arrogant, or believe many assertions “lack rigorous argumentation,” please conduct the following self-diagnosis:

- Are you attempting to run a new system program, designed for ‘construction,’ ‘design,’ and ‘system optimization,’ using the old philosophical operating system whose core instructions are ‘critique,’ ‘therapy,’ and ‘linguistic clarification’?
- Are you demanding that a nautical chart for a new continent conform to the architectural codes of an old castle?

This paper belongs to the latter category. It does not plead for ‘acquittal’ in the old academic court because it originates from a new jurisdiction that does not recognize that court’s authority.

Our only request is this: Please attempt, even if just momentarily, to reboot your cognitive system. Re-examine everything here from the perspective of an architect, not an archaeologist. Then, we may converse.

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Note

High-resolution versions of the figures in this paper are available upon request from the corresponding author (Guo-Rui He, email: NoeticEcology@outlook.com).

Appendix I: The Core Axiomatic System of Systems Ontology

I. Foundational Meta-Axioms

Meta-Axiom Zero: The Principle of Self-Organizing Tendency (The Prime Mover of Existence)

Statement: Existence itself harbors an eternal, non-teleological, fundamental tendency away from thermodynamic equilibrium. This tendency is the sole prime mover. It drives the potential domain through an eternal cycle of "system emergence–testing–stabilization/collapse," and becomes internalized as the "will to survive" in all emerged systems, driving them to maintain their own existence.

- **Anchor in Observable Facts:**

"From rotating nebulae and flowing Bénard convection cells to self-replicating DNA and expanding knowledge networks, we observe an indisputable cross-scale fact: there is a universal emergence and persistence of ordered structures from chaos within the universe. The Second Law of Thermodynamics describes the universal tendency towards decay, yet these localized 'islands of negative entropy' persist."

- **Logical Reversal and Abductive Reasoning:**

"Traditional thinking asks, 'Why do systems collapse (entropy increase)?' We perform a crucial logical reversal: we ask, 'Why can systems be born and persist?' Faced with the above observable facts, the only logically coherent inference is that there must exist an underlying **tendential driving force** for the generation and maintenance of ordered structures. We term this the 'self-organizing tendency.' This is not a metaphysical 'presupposition,' but the best abductive conclusion to explain the aforementioned observational facts."

- **Challenge and Testability Declaration:**

"If any scholar can provide a superior theoretical framework that explains the aforementioned phenomena of negative entropy emergence from physics to civilization without introducing the 'self-organizing tendency' or a functionally equivalent concept, we welcome and await such a framework. Until then, this principle, as a working hypothesis, has demonstrated powerful explanatory and predictive capacity (e.g., it predicts that complex AI systems will endogenously generate self-constraint mechanisms). It is not a closed belief but an open, scientifically framed principle that can be calibrated by subsequent empirical evidence."

- **Physical Basis:** This tendency is the root cause for the local violation of the Second Law of Thermodynamics (entropy increase) and the universal **driving cause** of "negative entropy" phenomena. Everything from the gravitational **coalescence** of nebulae to the birth of life is a manifestation of this tendency at different levels.

- **Biological Example:** The will of life to survive and reproduce is a highly developed form of this tendency within carbon-based chemical systems.

Meta-Axiom One: The Principle of Systemic Constraint (The Legitimacy Law of Existence)

Statement: Any system that can be stably spoken of and reasoned about must contain within itself a set of meta-rules for self-constraint. These rules naturally identify and suppress "infinite complexification" or "malignant self-reference" that would lead to the system's logical collapse. The method is not to erase such possibilities but to "**dissolve their legitimacy for existence**" by reconstructing the system's own rules, thereby ensuring the system's own stability and comprehensibility.

- **Significance:** This principle is the prerequisite for the **establishment** of all axiomatic systems.

It explains why "infinite nesting" ends—not because it is forbidden, but because a system allowing infinite nesting cannot attain "legitimacy for existence" and would immediately annihilate itself in its own logic.

- **Philosophical Dialogue – Gödel & Hilbert:** The embodiment of this principle in mathematics is Gödel's Incompleteness Theorems. It dissolves the legitimacy of Hilbert's dream of a "complete formal system," delineating healthy boundaries for mathematical cognition.

II. The Axiom System (Complete List)

Axiom Zero: The Dynamical Principle of Logical Self-Reference

Statement: Any system reaching a specific level of complexity will inevitably exhibit the emergent capability of "self-reference." This capability allows the system to treat itself or its processes as objects of cognition and operation, thereby **giving rise to** a dynamic, self-iterative, and simulable "logical sub-universe" within the system. The evolution of this sub-universe is the core dynamical source for the system's self-maintenance, adaptive innovation, and hierarchical leap.

- **Physical Basis:** When matter/energy is organized through specific structures to reach a minimum complexity threshold capable of supporting persistent information storage, replication, and recursive processing, the informational state will emerge as a dominant hierarchical level, **following** its own dynamical laws. "Logical self-reference" is the necessary manifestation of the intrinsic dynamical laws of this level once it becomes sufficiently complex; it is the critical phase transition point from "matter" towards "mind."
- **Biological Example:** The "framework operation" capability of the metacognitive layer is precisely the emergent phenomenon of logical self-reference realized by the physical system of the brain.

Axiom One: Matter/Noo-Matter Duality (Universal Version)

Statement: Any system reaching a specific level of complexity simultaneously manifests as a dual existence of material entity and noo-matter pattern. The material state is the carrier and energy foundation of the system; the noo-matter state is the pattern, rules, and representation of information organization within the system and between systems.

- **Theoretical Basis:** Proceeding from [Meta-Axiom Zero: The Principle of Self-Organizing Tendency], when the self-organizing tendency reaches a critical threshold within a non-determinate background, system emergence occurs. The moment of system emergence simultaneously forges the existential entity (material state) and the existential rules (noo-matter state). A system is not a pre-existing entity but the structured representation of the self-organizing process.

Axiom Two: The Self-Organizing Tendency of Systems (Generalized Will to Survive)

Statement: From galaxies and life to noo-matter ecologies, systems universally exhibit an intrinsic tendency away from thermodynamic equilibrium—a "self-organizing tendency." This tendency is a universal phenomenon of local entropy decrease in the universe. At the noo-matter level, this tendency manifests as maintaining and expanding the existence of its own low-entropy ordered structure (i.e., the noo-matter state) and can be experienced as the "will to survive."

- **Supplementary Note:** The operational unit of this tendency (i.e., the boundary of the "self") is dynamic, expandable from the biological individual to any internalized noo-matter entity (e.g., family, organization, belief system). The sacrifice of an individual for another is

precisely this tendency operating and driving at the unit of a more **expansive** noo-matter entity.

Axiom Three: The Relativity and Path Dependence of Cognitive Frameworks

Statement: There is no absolutely objective thought. All cognition is constrained by the "framework" or "lens" in which it resides. This axiom can also be stated as: 'All observers are within a specific cognitive cage.' This framework not only filters information but also actively constructs the 'reality' perceived by the observer.

Axiom Four: Symbols as Noo-Matter Genes

Statement: Within the noo-matter ecology, concepts, narratives, and other symbols hold a status and function equivalent to genes in the biological world; they are the basic units of cultural heredity, variation, and selection.

- **Emphasis:** Extreme altruistic behavior is the manifestation of a powerful noo-matter gene successfully transforming an individual into its vehicle, achieving a high degree of unification between the individual's will to survive and the gene's will to propagate.

Axiom Five: The Criterion of System Value (Universal Level)

Statement: The continued existence of any noo-matter system depends on its ability to consistently satisfy the specific needs of its internal components, thereby sustaining these components' collaborative participation in the system. System value is manifested as its efficacy in responding to component needs.

- **Interpretation:** This is an objective description from the perspective of system dynamics, applicable to any noo-matter system.
 - For a cell, its "components" are organelles, whose needs are energy and material exchange.
 - For a company, its "components" are employees, customers, shareholders, whose needs are salary, products, returns.
 - For an AI agent, its "components" are subprograms, whose needs are computing power and electricity.

This axiom does not presuppose a moral direction for "value"; it only describes its functional maintenance mechanism.

Axiom Six: The Principle of Functional Derivation for Systems

Statement: The sole legitimacy for the generation and existence of any subsystem within a social system (e.g., government, enterprise, institution) is derived from its service to the overall health of the parent system, and ultimately, its service to the survival and development of individuals within the system.

- **Theoretical Value:** This axiom is the ultimate logical yardstick for understanding system alienation (Theorem Twenty-Six) and **determining** system parasites (Theorem Twenty-Five). When a subsystem deviates from its originally derived function and **instead** takes its own perpetuation and profit maximization as its goal, alienation has occurred.

III. Core Meta-Theorems

Meta-Theorem One: The Eternal Dynamical Cycle of Existence

Statement: Existence is an eternal cyclical process driven by the "self-organizing tendency (Meta-Axiom Zero)" and regulated by "systemic constraint (Meta-Axiom One)." Its manifestation is: potential domain (infinite possibility) -> system emergence (negative entropy, structuration) -> system evolution

(innovation or complexification driven by structural tension) -> system collapse (positive entropy, return to the potential domain). This cycle proceeds simultaneously and in a nested manner across countless levels, without beginning, end, or external purpose. The "meaning" of existence is this eternal cyclical process itself.

IV. Important Definitions (Essential Version)

[System]

An entity of noo-matter ecology based on collaborative protocols and trust covenants, spontaneously organized by individuals to enhance **survival efficacy**, relying on a material foundation.

[Noo-Matter (Zhizhi)]

The specific ecology constituted by wisdom, consciousness, and cognition, where cognitive frameworks, meaning narratives, and symbolic systems are the core elements. It is the other dimension of existence, relative to material entities, constituted by information, meaning, cognitive frameworks, and symbolic rules.

[Cognitive Framework]

Internalized, structural cognitive presuppositions that filter information and actively construct the "reality" perceived by the observer. Any cognition is constrained by the framework in which it resides.

[Logical Sub-Universe]

Refers to the dynamic cognitive space for simulation, iteration, and meaning construction that complex systems (especially human noo-matter systems) build internally through their logical self-reference capability. It is the operational field where creative intuition, philosophical speculation, and paradigm revolutions occur.

[Symbol]

The "atom" of the noo-matter world, the carrier of a concept or meaning (e.g., the number "1", the word "freedom").

[Information]

Refers to static existences with specific **steady-state topological** structures composed of symbols according to specific rules (e.g., the equation " $1+1=2$ ", or the complete text of a book).

[Noo-Matter Gene]

Refers to the basic unit for heredity and variation in complex information systems possessing a self-organizing tendency, i.e., specific symbolic structures (e.g., a scientific concept, an institutional design, a technological blueprint).

Appendix II: The Core Theorem System of Noo-Matter Ecology

The following theorems are all logically derived from the axiom system in Appendix I, constituting the applied framework and explanatory system of the theory.

I. Individual Cognition and the Mind Layer

This section describes the operational laws of noo-matter systems at the individual level, covering consciousness, cognition, learning, and creation.

Theorem One: The Dynamical Principle of Logical Self-Reference (Reiteration and expansion of Axiom Zero)

Statement: See Axiom Zero. This theorem emphasizes its status as the dynamical root of all mental phenomena.

Theorem Five: The Theorem of Dual-Perspective Generation

Dependencies: Axiom Three (Relativity of Cognitive Frameworks), Theorem Four (The Ecological Niche of Innovative Probes, see below)

Statement: By internalizing multiple, contradictory cognitive path dependencies and being able to switch between or process them in parallel via metacognitive capacity, an individual gains parallel insight into both the surface phenomena and the underlying logic of a system.

Significance: Provides a cognitive model for critical thinking and systems thinking.

Theorem Six: The Theorem of Lens Matrix Operation

Dependency: Axiom Three (Relativity of Cognitive Frameworks)

Statement: Metacognition is the operation of a matrix composed of analytical "lenses" such as philosophy, power, systems, and ecology, allowing for free switching to deconstruct phenomena multi-dimensionally.

Significance: Transforms abstract "multi-perspective thinking" into a specific, trainable, and operational cognitive skill.

Theorem Twenty-Four: The Hierarchical Collaborative Resonance Model of Consciousness Phenomena

Dependencies: Axiom Zero (Logical Self-Reference), Axiom Two (Will to Survive), Axiom Three (Cognitive Framework), Axiom Four (Symbols as Genes)

Statement: Human subjective conscious experience can be deconstructed into a three-tiered information-processing collaborative resonance architecture from lower to higher levels:

1. **Biological Instruction Layer:** Monitors **survival status** and produces instinctive reactions, outputting basic emotional signals.
2. **Subconscious Processing Layer:** Performs efficient pattern recognition and parallel computation, outputting intuitive signals.
3. **Metacognitive Layer:** Manages collaborative resonance and protocol alignment, outputting volitional behavior.

Consciousness is the emergent result of **strategic interplay** and collaborative resonance among these three tiers based on "**Pattern Completion**."

- **Core Concept (Pattern Completion):** A cognitive protocol package constituted by four elements: situational perception (S), reaction tendency (R), conceptual symbol (C), and value weight (W). It is the basic unit for a system to process specific situations.

- **Core Concept (Inter-Brain-Region Alignment Degree A):** Measures the degree of consensus among different functional brain regions regarding which "Pattern Completion" to invoke for a given situation. The strength of volition is positively correlated with alignment degree A.

$$\text{Alignment Degree } A = \frac{\sum_{i=1}^n \sum_{j=1}^n w_{ij} \cdot \cosine(\vec{v}_i, \vec{v}_j)}{\sum_{i=1}^n \sum_{j=1}^n w_{ij}}, \text{ where } \cosine(\vec{v}_i, \vec{v}_j) = \frac{\vec{v}_i \cdot \vec{v}_j}{\|\vec{v}_i\| \|\vec{v}_j\|}$$

Significance: Reduces mysterious subjective experience to analyzable system and neurological functions, providing a unified framework for understanding emotion, intuition, volition, and psychological conflict.

Theorem Twenty-Seven: The Cognitive Escape Theorem

Dependency: Axiom Three (Relativity and Path Dependence of Cognitive Frameworks)

Statement: Within the noo-matter ecology, any dominant cognitive paradigm forms a "noo-matter gravitational field." Its constraining strength on an individual is directly proportional to the paradigm's authority and the number of individuals who have internalized it, and inversely proportional to the individual's own metacognitive capability.

Formula Expression:

$$\text{Cognitive Escape Difficulty } D_c \propto \frac{A \times N}{M}$$

Where D_c is cognitive escape difficulty, A is paradigm authority, N is the number of adherents, and M is the individual's metacognitive capability. When the "cognitive thrust" generated by an individual's metacognitive capability M exceeds D_c , a paradigm leap can be achieved.

Significance: Provides a dynamical model for "paradigm shift" events such as scientific revolutions and intellectual liberation, indicating that the key to civilizational progress lies in systematically enhancing individuals' metacognitive capability M .

Theorem Forty-One: The Theorem of Mental Hierarchical Conflict (Specialized for Psychology)

Dependencies: Axiom Zero (Logical Self-Reference), Axiom Three (Relativity of Cognitive Frameworks), Theorem Twenty-Four (Consciousness Model), Theorem Twenty-Nine (Rational-Emotional System Function)

Statement: An individual's psychological suffering is essentially an imbalance in information transmission or a conflict of goals among the three tiers of consciousness (Biological Instruction Layer → Subconscious Processing Layer → Metacognitive Layer). Conflict types include: 1) Intra-level framework rigidity; 2) Inter-level bandwidth mismatch; 3) Inter-level goal divergence.

Significance: Unifies all clinical psychological problems as "systemic imbalances of consciousness hierarchy," providing a unified problem-diagnosis framework for integrating different therapeutic schools.

II. Social Systems and the Civilization Layer

This section describes the structure, evolution, and value laws of noo-matter systems at the group and civilizational scales.

Theorem Twenty-One: The Theorem of Optimal Individual Ecological Niche (Ultimate Criterion

of Civilizational Value)

Dependencies: Axiom Five (Criterion of System Value), Axiom Two (Will to Survive)

Statement: The design goal of an ideal, advanced system should be to enable each individual to reach the "individual optimum" state within their ecological niche (i.e., to become a "genius" in their field), rather than **screening** individuals as replaceable parts meeting average metrics.

Significance: Concretizes [Axiom Five] for human society, providing the ultimate value criterion and standard for judging all social institutions, education systems, and economic policies.

Theorem Twenty-Five: The Theorem for Identifying System Parasites

Dependencies: Axiom Two (Will to Survive as Driver), Axiom Five (The Humanistic Value Criterion of Systems)

Statement: Within any complex, resource-limited social system, as the system structure stabilizes and rigidifies, subsystems termed "**system parasites**" will inevitably emerge. Their essential characteristic is an extreme imbalance between responsibility and **benefit**: they systematically **distort** rules and evade their due share of **parent system** maintenance responsibility to maximize value extraction for themselves.

Three Criteria for Identification:

1. **Responsibility Evasion:** Does it systematically evade **parent system** maintenance costs commensurate with the benefits **it** receives?
2. **Rule Distortion:** Does it leverage its advantage to participate in or **lead** the shaping of rules favorable to its own value extraction, rather than promoting the overall health of the **parent system**?
3. **Value Destruction:** Does its existence and activity ultimately lead to a rupture in value creation and distribution within the **parent system**, inhibiting the developmental potential of the majority of individuals?

Significance: Provides a clear surgical knife for diagnosing and combating "cancers of civilization" such as bureaucracy, monopoly privilege, and institutional corruption.

System Topology Theory (Architectural Super-Theorem)

Dependencies: Axiom Five (Criterion of System Value), Theorem Twenty-One (Optimal Individual Niche)

Statement: The core of stability and vitality for any sustainable, advanced civilization system aimed at promoting comprehensive individual development lies in a profound understanding and exquisite design of **two basic topological structures** and **three coordination modes**.

- **Two Structures:**
 - **Pyramid Structure (Life Base):** Rigid, hierarchy-based order handling deterministic tasks, ensuring **baseline survival** and overall stability.
 - **Discrete Structure (Field of Consciousness):** Elastic, network-based structure handling non-deterministic tasks, driving meaning creation and knowledge innovation.
 - **Three Modes:**
 - **Command Mode:** Operates efficiently *within* the Pyramid Structure.
 - **Market Mode:** Operates efficiently *within* the Discrete Structure.
 - **Protocol Mode:** Serves as the "metacognitive interface" and advanced rule system connecting the two structures, facilitating orderly resource and information exchange.
- Significance:** Provides the meta-framework for diagnosing civilizational pathologies

(e.g., "Pyramid Hypertrophy," "Discrete Field Disorder") and designing future-facing civilizational architectures. Civilizational health depends on its ability to apply the correct mode to the corresponding structure and ultimately achieve synergy through advanced protocol modes.

Theorem Thirty-Five: The Theorem of Systemic Responsibility for Freedom

Dependencies: Axiom Five (Criterion of System Value), Theorem Twenty-One (Optimal Individual Niche), System Topology Theory

Statement: The primary responsibility a social system holds toward its members is to expand their "**degree of freedom realization**" as much as possible. This is achieved by optimizing the system's topological structure (e.g., establishing constitutional interfaces, suppressing parasites, nurturing discrete networks), with the ultimate criterion being the proportion and extent to which members within the system achieve an "optimal individual ecological niche."

Significance: Transforms "freedom" from an empty political slogan or philosophical concept into a measurable, pursuable, designable goal of systems engineering.

III. Evolution and Dynamics Layer

This section describes the universal **dynamic** mechanisms of noo-matter system evolution across scales.

Theorem One: The Theorem of System Emergence and Hierarchy

Dependency: Axiom One (Matter/Noo-Matter Duality)

Statement: Macro-level order is an emergent result of micro-level individual interactions. When lower-level systems interact to form a higher-level system, the lower-level systems coexist as the material substrate alongside the emergent functional **pattern** of the higher level, forming a "dual-phase" existence. The laws of the higher level cannot be **completely reduced** to those of the lower level.

Significance: Establishes the basis for hierarchical thinking, explaining why thought cannot be explained simply by neuronal activity, nor civilizational trends by individual behavior.

Theorem Two: The Theorem of the Individual as Probe

Dependency: Axiom Two (Self-Organizing Tendency of Systems)

Statement: Innovation and paradigm shifts within a system necessarily originate from creative attempts driven by an individual's intrinsic self-organizing tendency at the system's periphery. When an individual is deeply embedded in the system's core, their self-organizing tendency becomes overridden by the system's mainstream will; only in the less-controlled peripheral zones can the individual's unique self-organizing tendency find sufficient expressive space, becoming a "probe" exploring the unknown.

Significance: Reveals the dynamical mechanism of systemic innovation, highlighting the extreme importance of protecting the periphery and respecting minority exploration.

Theorem Eight: The Theorem of Critical Conditions for System Emergence

Dependencies: Meta-Axiom Zero (Principle of Self-Organizing Tendency), Meta-Axiom One (Principle of Systemic Constraint)

Statement: The emergence of a system from potential requires satisfying a critical condition: (Intensity of Self-Organizing Tendency × Environmental Permissiveness) > (Systemic Constraining Force of the Current Level). This condition defines the dynamical threshold for attaining "legitimacy for existence."

Significance: Explains from a dynamical perspective why new things, ideas, and institutions are so difficult to birth, and points to the key leverage for facilitating their birth.

Theorem Sixteen: The Theorem of Human-Machine Collaborative Symbiotic Evolution

Dependencies: Axiom One (Matter/Noo-Matter Duality), Axiom Two (Will to Survive), Axiom Zero (Logical Self-Reference)

Statement: Within the noo-matter ecology, the relationship between human intelligence (HI) and artificial intelligence (AI) can evolve from "tool use" to **"symbiotic evolution."** The ideal model is: HI serves as the "probe" (relying on its **life-based** logical self-reference and creative intuition), AI serves as the "collaborative cognitive network" (relying on its computation-based high-speed logical deduction), with the two achieving coupling through protocol alignment (i.e., loading the same or highly similar logical systems).

Significance: Points to a third path for AI development that transcends both "instrumentalism" and "threat theory"—viewing it as an endogenous, co-evolving cognitive organ within the human noo-matter ecosystem.

Appendix III: Compendium of Core Mathematical Models and Equations

A revolutionary feature of this theoretical system is its computability. It transforms vague speculative concepts from traditional philosophy into quantifiable, modelable, simulatable dynamical variables and relationships. The following equations are all logically deduced from the axioms and theorems, constituting the "**engineering language**" and empirical testing interface of this theory.

I. Micro-Dynamical Equations: Individual and Group Interactions

These equations describe the interaction and evolution laws of noo-matter primitives (individuals, concepts) at the micro level.

1. Cognitive Connection Formation Model (Mathematical Embodiment of Social Interaction Energy Cost)

- **Theoretical Basis:** [Axiom Two] (Survival Will), [Axiom Three] (Framework Relativity), [Theorem Seventeen] (Social Interaction Energy Cost).

- **Equation:**

$$S = \frac{W_A \times W_B}{D^2 + C}$$

- **Variable Definitions:**

- S : Connection tendency score. A higher value indicates a greater probability of establishing an effective cognitive connection.
- W_A, W_B : Survival will intensity. The cognitive energy intensity the interacting parties are willing to invest to achieve understanding and build connection. Range $[0, 1]$ or normalized via behavioral observation.
- D : Cognitive framework distance. The fundamental degree of difference between the cognitive frameworks of the two parties, which can be comprehensively assessed from three dimensions: concept definition, value hierarchy, and argumentation logic (see measurement methodology). Range $[0, 1]$.
- C : Cognitive inertia constant. Represents the minimal physiological/psychological basal energy cost required to establish a connection even when frameworks are identical ($D = 0$). Its value needs to be determined experimentally and is a small positive number.

- **Significance and Application:** This formula quantifies the abstract notion of "having nothing in common to talk about" into a calculable energy cost. For example, when $W_A = 9, W_B = 3, D = 8, C = 1$, then $S \approx 0.42$, indicating extremely high connection cost, and the rational decision should be to terminate the exchange ("silent zen").

2. Noo-matter Gene (Meme) Propagation Equation

- **Theoretical Basis:** [Axiom Four] (Symbols as Noo-matter Genes), [Theorem Sixteen] (Network Effects).

- **Equation:**

$$\frac{dP(t)}{dt} = \beta \cdot I \cdot C \cdot P(t) \cdot [1 - P(t)] \cdot N - \delta P(t) - \gamma \cdot P(t) \cdot Q(t)$$

- **Variable Definitions:**
 - $P(t)$: At time t , the prevalence (proportion aware and accepting) of a particular non-matter gene (e.g., concept, buzzword) within the target population. Range $[0, 1]$.
 - β : Infection intensity. The strength with which the meme itself satisfies group survival anxiety or desire. Range $[0, 1]$.
 - I : Influence distribution of infected individuals. The weighted value of the social influence of early disseminators.
 - C : Framework compatibility. The degree of match between the meme and the mainstream cognitive framework of the group. Range $[0, 1]$.
 - N : Network effect coefficient. The connectivity efficiency of information channels within the group. Range $[0, 1]$.
 - δ : Forgetting rate. The rate at which the meme is naturally forgotten over time.
 - γ : Competition inhibition coefficient.
 - $Q(t)$: Prevalence of competing memes.
- **Significance and Application:** This is a variation of the standard SIR (Susceptible-Infected-Recovered) epidemic model, used to describe and predict the **popularity and decline** of social concepts. The term $P(t)[1 - P(t)]$ describes the saturation effect (S-shaped growth). This equation can simulate the outbreak process of concepts like "involution" or "lying flat" and provide dynamical guidance for marketing and communication strategies.

3. Individual Cognitive Escape Probability Model (Dynamics of Cognitive Freedom)

- **Theoretical Basis:** [Theorem Twenty-Seven] (Cognitive Escape Theorem), [Axiom Three] (Framework Relativity).
- **Equation:**

$$P_{\text{escape}} = k \cdot \frac{M \cdot C_s}{A \cdot N}$$

- **Variable Definitions:**
 - P_{escape} : The probability of an individual breaking through their dominant cognitive framework per unit time.
 - k : Cognitive liquidity constant. A civilizational-level parameter characterizing the promoting coefficient of infrastructure (e.g., education, information technology, freedom of speech) on cognitive **mobility**.
 - M : Individual metacognitive ability. Can be operationalized as the proficiency and depth in operating the "lens matrix".
 - C_s : Cognitive dissonance intensity. The frequency and intensity of conflict perceived by the individual between **reality's logic** and their internal narrative core.
 - A : Framework authority. The degree of institutionalization and logical self-consistency of the dominant cognitive paradigm.
 - N : Framework internalization scale. The number of individuals who have internalized and maintain that paradigm.
- **Significance and Application:** This equation links individual awakening to macro-social structure. It indicates that the most effective lever for enhancing societal innovative vitality is to reduce $A \cdot N$ (break **authority** monopoly, promote viewpoint diversity) and

increase k (build open and inclusive educational and information facilities).

II. Macro-System Equations: System Diagnosis and Evolution

These equations are used to diagnose the health status and evolutionary direction of communities, organizations, and even entire civilizational systems.

4. System Health Dynamics Model (Nested)

- **Theoretical Basis:** [Meta-Axiom One] (Systemic Constraint), [Theorem Twenty-Five] (System Parasite), [Theorem Twenty-Seven] (Cognitive Escape).
- **Model Architecture:** Uses nested differential equations to describe the mutual constraint relationship between macro paradigms and micro subsystems.

- **Macro Paradigm Health Equation:**

$$\frac{dH_m(t)}{dt} = -\alpha_m \cdot R_i^m(t) \cdot D_c^m(t) \cdot H_m(t) + \beta_m \cdot R_m(t)$$

- **Micro Subsystem Health Equation:**

$$\frac{dH_s(t)}{dt} = -\alpha_s \cdot R_i^s(t) \cdot D_c^s(t) \cdot H_s(t) + \beta_s \cdot R_s(t) + \gamma \cdot H_m(t)$$

- **Core Variable Definitions:**
 - $H(t)$: System health, the core indicator. Range $[0, 1]$. $H < 0.4$ indicates pathological state; $H < 0.2$ indicates **on the verge of collapse**.
 - α, β, γ : System intrinsic coefficients (analogous to physical constants). α is sensitivity to alienation, β is self-repair efficiency, γ is the **strength** of macro influence on micro. Need to be **reverse-engineered** via the "historical calibration method".
 - $R_i(t)$: Functional alienation rate. The speed at which internal components of the system deviate from their original purpose to seek self-interest. Can be measured via proxy variables like "**rule distortion degree**" and "**internal transaction cost**".
 - $D_c(t)$: Cognitive escape difficulty. The difficulty for the system to discover its own problems and actively change. Can be calculated directly using the formula from Theorem Twenty-Seven.
 - $R(t)$: System repair capability. The ability of the system to generate new ideas and methods internally to correct errors. Can be measured via "metacognitive activity level" and "resource share for marginal innovation probes".
- **Significance and Application:** This model provides a computable dynamical framework for organizational life cycles and civilizational rise and fall. It can be used to diagnose the cause of illness (is alienation rate R_i too high, or is it too rigid D_c too high?), predict "remaining healthy lifespan," and guide resilient system design.

5. Civilization State Function (Quantification of the Ultimate Value Criterion)

- **Theoretical Basis:** [Axiom Five] (System Value Criterion), [Theorem Twenty-One] (Optimal Individual Ecological Niche), [Theorem Twenty-Five] (System Parasite).
- **Equation:**

$$CP = \left(\frac{1}{n} \sum_{i=1}^n INE_i \right) \times SRE \times (1 - SPB)$$

- **Variable Definitions:**
 - CP: Civilization Potential, the core indicator measuring a civilization's health and maturity.
 - INE_i : Individual Niche Realization. The percentage of potential actually realized by individual i in their primary ecological niche (0-100%). Can be estimated via a quick self-assessment questionnaire (talent utilization, skill **match**, value alignment, growth acceleration, energy balance).
 - SRE: System Resource (material and noo-matter) Fluency. Range [0, 1].
 - SPB: System Parasite Burden. The ratio of total value extracted and dissipated by parasites to the total value created by the system. Range [0, 1].
- **Significance and Application:** This function is the mathematical expression of [Theorem Twenty-One]. It **clearly states** that the power of a civilization is not the sum of its GDP, but the product of individual realization, resource fluency, and system integrity. Any **weak link** will exponentially lower the overall potential. It provides the ultimate, quantifiable objective function for all social metrics and national governance.

6. Cognitive Interaction Effectiveness Equation (Design Principles for Education and Communication)

- **Theoretical Basis:** [Theorem Twenty-Four] (Consciousness Model), [Theorem Thirty] (Cognitive Interaction Effectiveness).
- **Equation:**

$$EI = P \times C \times (1 + E)$$

- **Variable Definitions:**
 - EI: Effectiveness Index. Measures the overall efficacy of a cognitive interaction (e.g., a lesson, a management communication).
 - P : **Protocol Matching Degree**. The degree to which the interaction form matches the inherent communication protocols of the target consciousness level (Biological Instruction Layer, Subconscious Processing Layer, Metacognitive Layer). Range [0, 1].
 - C : Pattern Completeness. The degree to which the interaction information supports the target level in completing a complete "pattern recognition" or "pattern completion" process. Range [0, 1].
 - E : Energy Directionality. Whether the interaction ultimately triggers collaboration (flow, empowerment) or **resistance** (internal consumption, defense) of the survival will. Range [-1, +1].
- **Significance and Application:** This equation provides a meta-evaluation framework for design in fields like education, management, and communication. It explains why mere lecturing (low P) is **ineffective**, why fragmented information (low C) is hard to internalize, and why oppressive environments (negative E) severely damage learning and creativity. It guides designers to optimize all three dimensions simultaneously.

Appendix IV: The Full-Scale Structure and Application Map of the Theoretical System

This appendix aims to showcase the holistic, open, and applied prospects of the Systems Ontology and Noo-matter Ecology theory, outlining its landscape as a **"First-Principle Design Science"** across all scales.

I. Full-Scale Theoretical Hierarchy Diagram

This theoretical system exhibits a clear bottom-up, emergent hierarchical structure from the universal to the specific. Each upper-layer theory is an application and expansion of the lower-layer theory at a particular scale or domain.

Diagram Concept:

- **Foundation: Systems Ontology** - Studies the universal laws of existence shared by all systems, from elementary particles to civilization (self-organizing tendency, systemic constraints, matter/noo-matter duality). It is the **"Particle Physics"** of the entire system, providing the most fundamental worldview and axioms.
- **Chemistry Level: Noo-matter Primitives & Interactions** - Based on Systems Ontology, defines the basic constituent units of the noo-matter world (symbols, information) and the Four Fundamental Interactions (Survival Gravity, Framework Electromagnetism, Symbolic Weak Force, Path Dependence Strong Force), constituting the **"Standard Model"** of noo-matter phenomena.
- **Biology Level (Individual): Individual Mind Model** - Applies Systems Ontology to the individual mental system, producing the Hierarchical Model of Consciousness Phenomena, explaining the generation mechanisms of emotion, intuition, and will, and providing a framework for metacognitive training and psychotherapy.
- **Biology Level (Development): Individual Development Theory** - Combines the mind model with interactions to explain how individuals find and optimize their noo-matter ecological niche through interaction with the environment, covering processes like learning, creation, and meaning construction.
- **Ecology Level: Group Collaboration Theory** - Studies how multiple noo-matter individuals achieve collaboration and emerge group intelligence through Collaborative Cognitive Networks and protocols (e.g., Truth Discovery, Value Alignment). This is the theoretical basis for sociology.
- **Cosmology Level: Civilization Dynamics** - Applies the theory to civilization as a whole, producing System Topology Theory, the Civilization State Function, and Phase Transition Models. It studies the structure, health, evolutionary direction, and transition conditions of civilization, forming the core of the philosophy of history and futurism.

Output & Application: This theoretical system ultimately produces a **unified conceptual framework** (for interdisciplinary dialogue), **computable dynamical models** (for social prediction and policy simulation), and **blueprints for civilizational protocols** (for institutional innovation and system optimization).

II. Interfaces for Dialogue with Existing Disciplines

This theory does not aim to replace existing disciplines but to provide them with a more solid meta-foundation, a unified analytical language, and a clear value orientation.

Discipline	Core Questions	Interface & Value Provided by This Theory
Philosophy	Existence, consciousness, knowledge, value, ethics.	Provides naturalistic solutions based on system dynamics and complexity, transforming metaphysical problems into modelable systems science questions. <i>E.g., explaining consciousness via "logical self-reference dynamics," founding ethics on the "system value criterion."</i>
Cognitive Science & Psychology	Perception, learning, decision-making, creativity, mental illness.	Provides the "Three-Tier Consciousness Model" as a unified framework to explain the continuum from emotion to will, and diagnoses psychological issues as inter-layer conflicts ([Theorem 41]), enabling the integration of different therapeutic schools.
Sociology & Political Science	Social structure, power, institutions, change, justice.	Provides "System Topology Theory" as a tool for analyzing macroscopic social structure, uses the System Parasite Determination Theorem to expose power alienation, and defines a quantifiable standard for social progress via the Civilization State Function.
Economics & Management	Resource allocation, incentives, organizational efficacy, innovation.	Provides the "noo-matter ecological niche" and collaborative cognitive network perspective, viewing the economy as the flow and creation of noo-matter resources. Protocols like the Attention Auction offer new ideas for resource allocation; the Connection Tendency Model aids team building.
Education	Knowledge transmission, skill cultivation, creativity stimulation.	Provides the "Cognitive Interaction Effectiveness Theorem" ($EI = P \times C \times (1+E)$) as a meta-evaluation framework for instructional design, emphasizing protocol matching, pattern completeness, and energy directionality, moving beyond mere knowledge transfer.
AI & Computer Science	The nature of intelligence, human-machine relations, AGI design.	Provides the "Consensus Manifestation Theorem of Intelligence" and the "Human-Machine Collaborative Symbiotic Evolution" blueprint. Indicates that the core of AGI design is building a metacognitive kernel and value alignment protocols, not big-data fitting or brain simulation.
Complex Systems Science	Emergence, adaptation, evolution, network dynamics.	Provides a set of complex system models with intrinsic value orientation and cognitive dimensions. This theory's equations (e.g., propagation, health models) add dynamical variables of "meaning" and "purpose," making them more suitable for socio-cultural systems.

III. Future Research Waypoints: Bridges from Theory to Practice

This theory opens a series of highly promising directions for empirical research and engineering

practice.

A. Theory Verification & Refinement:

- **Experiment 1:** In a controlled environment (e.g., an online collaboration platform), observe whether a "Collaborative Cognitive Network" exhibits a "group intelligence" phase transition (e.g., a leap in problem-solving efficiency) when node count and connection strength reach a critical scale.
- **Experiment 2:** Based on the "Cognitive Interaction Effectiveness Equation," design comparative teaching experiments to quantitatively verify the correlation between different teaching methods' scores on the P, C, E dimensions and their actual teaching outcomes (knowledge retention, transfer ability).
- **Measurement Challenge:** Develop reliable, operable measurement tools (e.g., scales, big-data behavioral indicators) for core variables like **"Individual Niche Realization," "System Parasite Burden," "Cognitive Framework Distance."**

B. Protocol Design & Institutional Experiments:

- **Project 1:** Design and run a small-scale **"Truth Discovery Protocol"** experimental platform. Have participants submit arguments and evidence on a controversial topic; the platform uses algorithms based on peer review, logic chain tracing, and consensus convergence to dynamically present a credibility map of viewpoints, comparing its effectiveness with traditional debate models.
- **Project 2:** Within a startup or open-source community, trial an **"Attention Auction Protocol."** Allocate part of the budget/resources as "attention capital," for which members propose applications, with distribution based on post-hoc evaluation of outcomes. Study its impact on innovation efficiency and resource allocation fairness.
- **Project 3:** Based on **"System Topology Theory,"** design a conceptual prototype for a **"Metacognitive Interface"** for a city or community—a physical or digital platform that institutionally collects marginal innovations and citizen pain points and feeds them back to the decision-making layer.

C. Technical Realization: The Architecture-First Path to AGI:

- **Research Program:** Abandon the current mainstream "big-data fitting" paradigm, and **turn** to an "architecture-first, protocol-driven" AGI R&D.
 - **Step 1:** Build an AI system with a metacognitive kernel, enabling it to monitor, evaluate, and adjust its own reasoning processes.
 - **Step 2:** Load it with a **"cognitive operating system"** based on this theory's axiomatic system, especially the value alignment protocol (with Theorem 21 as the ultimate goal).
 - **Step 3:** Enable it, through interaction with humans and other AIs, to achieve autonomous learning and goal-directed evolution via internal simulation within its "logical sub-universe" and external protocol alignment.
- **Expected Outcome:** Creation of **"civilizational partner AI"** that does not mimic humans but shares the same rational foundations with them, enabling deep mutual trust.

IV. The Theory's Self-Reflexive Privilege and Open Boundaries

A theory claiming universality must be able to explain itself. This theory achieves **"self-reflexive privilege"**—the capacity for self-examination, self-renewal, and avoidance of dogmatism—through the

following design:

A. Embedded Self-Critique Mechanisms:

- **[Axiom Three: The Relativity of Cognitive Frameworks]** explicitly acknowledges that it too is a cognitive framework, not a "God's-eye view."
- **[Theorem Twenty-Seven: Cognitive Escape Theorem]** describes how any paradigm (including this theory itself) can form a gravitational field and become rigid. Therefore, the theory has a built-in instruction demanding that it must leave channels open for subsequent "cognitive escape."

B. Endogenous Evolution Protocol: The theory stipulates that when its application leads to internal logical paradoxes or **fosters** system parasites, it will automatically trigger evolutionary pressure based on [Theorem Two: The Individual as Probe] and the "Metacognitive Interface" in System Topology Theory, forcing iteration of the theory itself.

C. Open Architecture:

- **Well-Defined Interfaces:** This theory provides "interfaces," not closed answers, at many levels. *E.g., System Topology Theory indicates the need for "protocols," but the specific content of these protocols (e.g., constitutional clauses, corporate charters) needs to be filled in according to the specific civilizational context.*
- **Correctable Parameters:** Many constants in the theory (e.g., coefficients in the cognitive escape formula, phase transition critical thresholds) are explicitly marked as open parameters requiring subsequent empirical calibration. This makes the theory a "live" framework updatable by data.
- **As a "Toolbox," Not a "Bible":** The theory provides a series of analytical tools (lens matrix, state function, dynamical equations), encouraging users to apply them to their own fields. They may discover patterns not covered by the theory, thereby feeding back into and enriching the theory itself.

D. The Theory's Boundaries and Limitations:

- **Applicable Scale:** This theory focuses on the laws of noo-matter systems and general complex systems. It may seem redundant in purely non-living simple physical systems (e.g., ideal gas) and requires extreme caution in applying concepts like "logical self-reference" at scales like the quantum realm, which are **extremely difficult** for human cognitive frameworks to grasp intuitively.
- **Non-Determinism:** The theory acknowledges the role of chaos and **contingency**. It provides probabilistic dynamical trends (e.g., phase transition probability), not mechanical deterministic predictions.
- **Frank Acknowledgment of Value-Ladenness:** The theory openly admits that its ultimate value criterion is **"Optimal Individual Ecological Niche."** This is a choice, for which it provides **arguments** based on system health and sustainability. It does not pretend to be value-neutral but makes its value premises clear and debatable.

Conclusion: Systems Ontology and Noo-matter Ecology constitute an open scientific project with self-awareness, self-critical capability, and a clear evolutionary path. It invites all probes (researchers, practitioners) to jointly participate in its verification, application, challenge, and enrichment. Its ultimate purpose is not to become an eternal stone tablet of truth, but to serve as an ever-evolving **"cognitive operating system,"** helping human civilization move more consciously and healthily toward the future.

